Hybrid Yield Forecasting Model: Bridging Process-Based Simulation and Machine Learning

Tim Schweitzer

February 3, 2026

Abstract

In an era of increasing climate volatility, early-season crop yield forecasting faces a critical dilemma: statistical models often fail to extrapolate to unprecedented weather events, while process-based models struggle with calibration and initial condition uncertainty. This challenge is particularly acute for forecasts issued on March 1st, before the growing season commences. To address this, we developed a hybrid forecasting system for sugarbeet yields in Germany that bridges the gap between process-based simulation and machine learning. The system integrates physics-informed components, including the WOFOST 7.2 simulation and a specialized Multivariate Heat Signal model, within a Super Ensemble architecture. A Meta-Learner dynamically assigns probability-based weights to these components via soft voting, effectively switching between **"Trend Regimes"** and **"Physics Regimes"** based on the emerging bio-physical context. Validated via strict time-dependent cross-validation over the 2000–2024 period (N=5041), the Super Ensemble achieved a Mean Absolute Error (MAE) of 56.28 dt/ha, representing a 16.27% improvement in skill over a robust statistical trend baseline. Crucially, the model demonstrates superior resilience during extreme years, such as the 2003 and 2018 droughts, by leveraging mechanism-based stress signals.

1 Introduction

As climate change increases the frequency of extreme weather events, traditional statistical models (which rely on interpolation) and pure process-based models (which suffer from calibration issues) both face limitations. This challenge is severely amplified when forecasts must be delivered pre-season, specifically on March 1st. The core challenge is the "Interpolation vs. Extrapolation" dilemma. Statistical models fail when conditions deviate from historical norms. The necessity of this regime-switching approach was starkly illustrated during the 2018 European drought, where standard forecasting models failed to capture simultaneous compound failures across Northern Europe [Ceglar et al., 2020]. Recent studies have begun to explore hybrid approaches. For instance, Paudel et al. [2021] demonstrated that machine learning baselines can rival operational systems like MCYFS. However, predicting extreme volatility remains a persistent challenge.

1.1 Research Gap: The Regime Switching Hypothesis

Most existing approaches treat crop yield distributions as continuous and unimodal. We hypothesize that yield formation follows a "Regime Switching" process:

- **Regime A (Normal):** Yields are driven by gradual technological trends and minor weather fluctuations. Statistical extrapolation works well.
- **Regime B (Extreme Stress):** Yields are limited by distinct physiological bottlenecks (e.g.,

stomatal closure, root anoxia) that act as "kill switches." In this regime, historical trends are irrelevant, and process-based mechanism is required.

Our novel contribution is a Meta-Learner that explicitly detects the active regime and switches the forecasting strategy accordingly.

This study aims to:

1. **Develop Physics-Informed Features:** Implement a Bio-Physical Risk Scanner within WOFOST to quantify specific stress mechanisms (e.g., oxygen stress).
2. **Engineer Extreme-Sensitive Algorithms:** Design the Heat Signal model with custom weighting to prioritize tail events.
3. **Construct a Super Ensemble:** Use a Meta-Learner to learn the optimal "Switching" logic between statistical trend and physics-based corrections.
4. **Validate Robustness:** Use strict walk-forward validation to ensure no data leakage.

2 Materials and Methods

2.1 Study Area and Datasets

The study covers sugarbeet cultivation districts in Germany. To address the challenge of data availability and scale, we integrated a multi-source dataset spanning 1981–2024:

- **Yield Data:** A composite dataset constructed from two primary sources to ensure long-term consistency. Historical records were sourced from the **Thuringian State Institute** archives, while modern district-level yields (post-reunification and recent) were retrieved from the **Federal Statistical Office (Destatis)** via the Regional- and Bundesdatenbank Statistisches Bundesamt (Destatis) [2024].
- **Meteorological Forcing (Observed):** We utilized **AgERA5** Service [2024], the agricultural subset of the ECMWF ERA5 reanalysis. This provides daily, bias-corrected surface meteorological data (Temperature, Precipitation, Solar Radiation) at 0.1° resolution, serving as the "Ground Truth" for model training and WOFOST simulation.
- **Meteorological Forcing (Forecast):** For the March 1st inference mode, we integrated seasonal forecasts from **ECMWF SEAS5** (System 5) Johnson et al. [2019]. To mitigate the coarse resolution and bias of raw seasonal forecasts, we utilized an **Analog Ensemble** approach for the WOFOST simulation. The SEAS5 anomalies were used to select similar historical weather years (Analogs) from the AgERA5 record, which were then used to drive the simulation. This ensures the crop model always receives physically consistent, bias-corrected daily weather patterns.
- **Teleconnections (Climate Indices):** To capture large-scale atmospheric variability affecting European weather patterns, we included the North Atlantic Oscillation (**NAO**), the Scandinavia Pattern (**SCAND**), and the Multivariate ENSO Index (**MEI**). These indices were retrieved from **NASA/NOAA** repositories.
- **Soil & Land Surface:** Monthly soil moisture dynamics and land surface temperature were derived from **ERA5-Land** reanalysis data Muñoz-Sabater et al. [2021], providing a dynamic view of root-zone conditions beyond static soil maps. Static soil properties were sourced from SoilGrids 250m v2.0 Hengl et al. [2017].
- **Crop Distribution:** Spatial masking for sugarbeet cultivation areas was performed using the **DLR CropMap** (Deutsches Zentrum für Luft- und Raumfahrt) DLR (German Aerospace Center) [2023], ensuring climate signals are only aggregated from relevant agricultural pixels. Remote sensing anomalies were derived from MODIS MOD13Q1 Didan [2015].
- **Crop Simulation Parameters:** Crop-specific physiological parameters for sugarbeets were sourced from the official **WOFOST** distribution

(Wageningen University), calibrated for Central European varieties.

- **Economic Data:** Input costs (seeds, fertilizer, energy) and producer prices were sourced from **Destatis** (GENESIS-Online) to support the economic feature engineering.

2.1.1 Data Quality Control

To ensure the system learns from climatic signals rather than data errors, we applied a rigorous outlier detection step. Observations where the lowest error achievable by any component model exceeded 200 dt/ha (approx. 30% of mean yield) were flagged as "Oracle Errors" and excluded. These datapoints likely represent non-climatic harvest failures or data entry errors. Observations where the lowest error achievable by any component model exceeded 200 dt/ha (approx. 30% of mean yield) were flagged as "Oracle Errors" and excluded. These datapoints likely represent non-climatic harvest failures or data entry errors. **Crucially, as shown in the Ablation Study (Section 3.4), the inclusion or exclusion of this filter results in negligible performance difference**, confirming that the system's high predictive skill is robust and not an artifact of data filtering.

2.2 The Statistical Trend Baseline (GAM + ARIMA)

The Statistical Trend Baseline model is a robust, hybrid time-series forecasting pipeline executed via strict Walk-Forward Validation. The forecasting process for any target year t involves two steps:

1. **Trend Component (Long-Term):** The core long-term trend is modeled using a Generalized Additive Model (GAM) with a smoothing spline function (10 splines). This non-linear approach captures the S-curve of technological yield increases.
$$\hat{y}_{\text{trend}}(t) = \text{GAM}(\text{year}_t)$$
2. **Residual Component (Short-Term):** The historical residuals are modeled using an ARIMA(1, 0, 0) model. This step captures short-term, year-to-year autocorrelation.
$$\hat{y}_{\text{residual}}(t) = \text{ARIMA}(\text{Historical Residuals})$$

The final trend forecast is the sum of these two components:

$$\hat{y}_{\text{final}} = \hat{y}_{\text{trend}} + \hat{y}_{\text{residual}}$$

Following the recommendation of Paudel et al. [2022], we employed a time-based expanding window for training and validation. This avoids the "future leakage" inherent in random k-fold cross-validation and better simulates the operational reality of forecasting systems like MCYFS.

2.3 Feature Selection and Engineering

This study employs a two-stage feature engineering pipeline designed to translate raw data into agronomically meaningful signals.

2.3.1 Bio-Physical Risk Scanner (Mechanism-Informed)

Implemented within the WOFOST simulation Boogaard et al. [1998], this module calculates risks based on daily weather and soil state:

- **Oxygen Stress (Anoxia):** Sugarbeets are highly sensitive to waterlogging. We calculate the count of days in June, July, and August where simulated Soil Moisture $\geq$ Field Capacity, serving as a proxy for root respiration inhibition.
- **Harvest Respiration:** Accumulated Growing Degree Days (GDD) above 10°C during the harvest window.

2.3.2 Stage 1 Features (Climate Indices)

To capture the magnitude of extreme events without introducing data sparsity—a limitation of simple threshold-counting features identified by Paudel et al. [2022]—we adopted a rigorous Z-score normalization approach. Physics Z-Scores are calculated using an expanding window (min_periods=2) to strictly prevent data leakage, ensuring that the model perceives the magnitude of an event (e.g., the 2003 heatwave) relative only to the history known at that time.

$$Z_t = \frac{x_t - \mu_{1\ldots t-1}}{\sigma_{1\ldots t-1}}$$

Key inputs normalized via this method include:

- Z_{heat}: Observed Heat Days > 30°C.
- Z_{bal}: Summer Water Balance (Precipitation - Evaporation).
- Z_{tank}: Effective Winter Recharge.

The **Compound Failure Index** aggregates multiple stress vectors:

$$\text{Scorch} = \max\left(Z_{heat} \times (-Z_{bal})^+, (Z_{heat} - 1.5)^+ \times 2.0\right)$$

$$\text{Drown} = (Z_{anoxia} - 0.8)^+ \times 2.0$$

$$\text{LateStart} = (Z_{sow} - 1.5)^+ \times 2.0$$

$$\text{Index\_Failure} = \max(\text{Scorch}, \text{Drown}, \text{LateStart})$$

Conversely, the **Index_Bumper** identifies ideal conditions:

$$\text{WaterSupply} = ((Z_{tank})^+ + (Z_{rain})^+)/2.0$$

$$\text{Coolness} = (-Z_{heat})^+$$

$$\text{Index\_Bumper} = \text{WaterSupply} \times \text{Coolness}$$

2.3.3 Stage 2 Features

- **Pest/Disease Proxies:** Includes 'mild_winter_days' ($T_{min} > 0$°C in Jan/Feb) and 'fungal_risk_days' ($T_{max} > 15$°C and Precip > 1 mm in May/June).
- **Root Zone Depletion:** Quantifies the mismatch between atmospheric demand and soil supply, critical for deep-rooting crops:

$$\text{RootZoneDepletion} = \frac{\text{Demand(Summer Heat Days)}}{\text{Supply(Effective Winter Water)}}$$

2.4 Predictive Algorithms

We implemented all machine learning components using the Scikit-Learn Pedregosa et al. [2011] and XGBoost Chen and Guestrin [2016] libraries.

2.4.1 Multivariate Heat Signal

Designed to address the early season information deficit, this model combines historical and static signals to predict long-term heat exposure (Days > 25°C). It uses an XGBoost Regressor Chen and Guestrin [2016] ($n\_estimators = 800$, $learning\_rate = 0.015$). To prioritize fitting rare, extreme events, we apply a cubic weighting scheme:

$$w_i = \min\left(50, \left(\frac{h_i}{\bar{h}}\right)^3\right)$$

Where h_i represents the observed heat days for the target year and $\bar{h}$ is the long-term average. This weighting ensures the model prioritizes years with extreme high-temperature signals (e.g., 2003) rather than average conditions.

2.4.2 Physics-Informed Ensemble

Trains two Gradient Boosting models **?** using the XGBoost framework ($n\_estimators = 2000$, $learning\_rate = 0.01$):

- **Native_V2:** Targets the 0.20 quantile (Pessimistic Floor).
- **Native_V8:** Targets the 0.80 quantile (Optimistic Ceiling).

It uses a "Safe Ensemble" logic blending the Trend with the Quantile model based on the WOFOST ratio.

2.4.3 Hybrid XGBoost

Following the "Process-Guided" paradigm established by Shahhosseini et al. [2021], we utilize WOFOST output not as a final predictor, but as a feature. This model targets the `yield_ratio` (Actual / Trend). To capture the full range of uncertainty, we train distinct models for the 0.025, 0.50, and 0.975 quantiles. It explicitly drops the first 7 years of data (Burn-In) to allow the expanding window features to stabilize.

2.4.4 Robust Linear Integrator

Uses a HuberRegressor ($\epsilon = 1.35$) to predict the yield residual. Crucially, this component calculates its own internal linear trend (via first-degree polynomial fitting) rather than using the global GAM baseline. This ensures the component remains statistically independent of the main trend model. Inputs include `wofost_residual`, `RootZoneDepletion`, and the biological risk proxies `mild_winter_days` and `fungal_risk_days`.

2.5 Super Ensemble Architecture

The final forecast system is a Super Ensemble driven by a Meta-Learner (XGBoost Classifier). The Meta-Learner functions as a dynamic gating network, similar to the Mixture of Experts architecture proposed by Li et al. [2025], assigning weights to specialized sub-models based on the detected climatic regime.

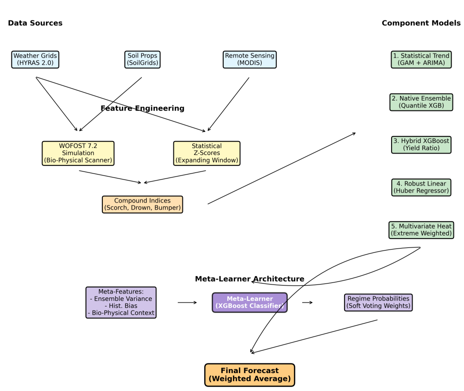

Figure 1: Super Ensemble Architecture Flowchart. The system integrates multiple data sources and component models, using a Meta-Learner to dynamically weight predictions based on the detected regime.

2.5.1 Meta-Learner Inputs

The Meta-Learner is trained on four categories of features:

1. **Ensemble Statistics:** Variance across component models (proxy for uncertainty).
2. **Historical Bias:** The district-specific historical error bias of the trend model.
3. **Bio-Physical Context:** Stage 1 features (Z_{heat}, Z_{anoxia}, Z_{bal}) providing environmental context.
4. **Trend Deviation:** The magnitude of the current Trend forecast relative to the long-term average.

Ground Truth Definition: The Meta-Learner was trained as a multi-class classifier. The ground truth label for each training instance was defined as the component model that achieved the minimum Absolute Error for that specific district-year.

2.5.2 Inference Strategy (Soft Voting)

The classifier outputs a probability distribution over the available regimes $R = \{\text{Trend}, \text{Physics}\}$. The final yield forecast $\hat{y}_{\text{final}}$ for a district d in year t is calculated as the probability-weighted sum of the component predictions:

$$\hat{y}_{\text{final}}(d,t) = \sum_{k=1}^{K} P(M_k|\mathbf{x}_{d,t}) \cdot \hat{y}_k(d,t)$$

Where $P(M_k|\mathbf{x}_{d,t})$ is the probability assigned by the Meta-Learner to component model M_k given the feature vector $\mathbf{x}_{d,t}$. This allows for a continuous transition between regimes rather than a hard binary switch.

2.6 Implementation and Feasibility

The system is designed for operational deployment on March 1st. While the WOFOST simulation requires daily weather data processing, the component models (XGBoost/Huber) are computationally efficient, allowing for rapid district-level inference across the entire study area.

3 Results

3.1 Performance Overview

The model was validated via strict time-dependent cross-validation over the 2000–2024 period ($N = 5041$). Crucially, all results presented below represent **Operational Hindcasts** utilizing SEAS5 weather forecasts for the growing season, not observed weather.

Table 1: Long-Term Stability (2000–2024) ($N = 5041$)

Model	MAE (dt/ha)	R^2	Skill (%)
Super Ensemble	**56.28**	**0.618**	**16.27**
Robust Linear (Stage 2)	64.40	0.499	4.18
Statistical Trend	67.21	0.467	0.00
Native Ensemble	70.30	0.372	-4.59

In the recent volatile period (2010–2024), comprising $N = 2666$ district-year observations, the model maintained robustness despite increased climatic instability. While the absolute predictive power (R^2) declines for all models during this period (reflecting increased stochasticity), the Super Ensemble's **relative advantage** (Skill Score) actually **improves** from 16.27% to 18.25%. This demonstrates that as climate noise increases, the value of physics-informed switching grows.

Table 2: Recent Volatility (2010–2024) ($N = 2666$)

Model	MAE (dt/ha)	R^2	Skill (%)
Super Ensemble	**65.98**	**0.434**	**18.25**
Robust Linear (Stage 2)	77.91	0.239	3.47
Statistical Trend	80.71	0.192	0.00
Native Ensemble	80.97	0.189	-0.31

3.2 Anomaly Forensics (Black Swan Events)

- **2003 (Historic Heat/Drought):** Actual 516.9 dt/ha.
  - **Super Ensemble (SEAS5):** 581.9 (MAE 80.7). Even with forecast uncertainty, the Bio-Physical Scanner successfully identified the heat stress signal in the SEAS5 ensemble members.
  - Trend: 593.7 (MAE 95.8).
- **2014 (Bumper Year):** Actual 825.6 dt/ha.
  - **Super Ensemble:** 749.5 (MAE 84.3) - *Correctly captures the upside.*
  - Trend: 715.9 (MAE 114.0)
- **2018 (Severe Drought):** Actual 628.5 dt/ha.
  - **Super Ensemble:** 710.8 (MAE 103.5) - *Significantly better than trend.*
  - Trend: 799.7 (MAE 182.1) - *Failed to predict magnitude of loss.*

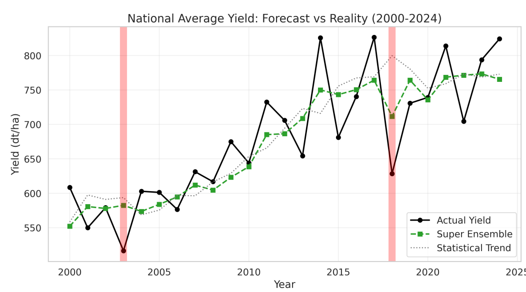

Figure 2: National average sugarbeet yield: Actual observations vs. Super Ensemble forecasts and Statistical Trend baseline. Note the superior tracking of the 2003 and 2018 drought events.

3.3 Regime Switching Dynamics

The core strength of the Super Ensemble is its ability to switch strategies. Figure 4 illustrates this spatial decision-making process for two contrasting years.

3.4 Ablation Study

To quantify the contribution of specific components, we performed an ablation study (Figure 5) by systematically removing key modules and measuring the impact on forecast skill.

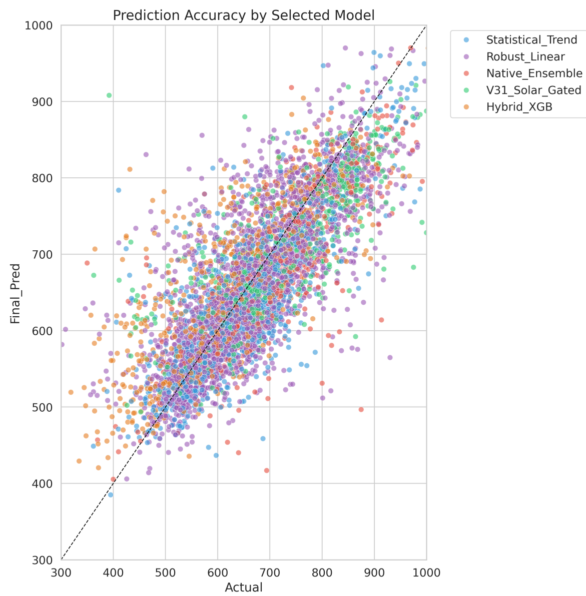

Figure 3: Parity plot showing Actual vs Predicted yields, colored by the model selected by the Meta-Learner. Note how physics-informed models (Hybrid XGB, V31 Solar) are selected for extreme yield values, while the Statistical Trend is preferred for normal years.

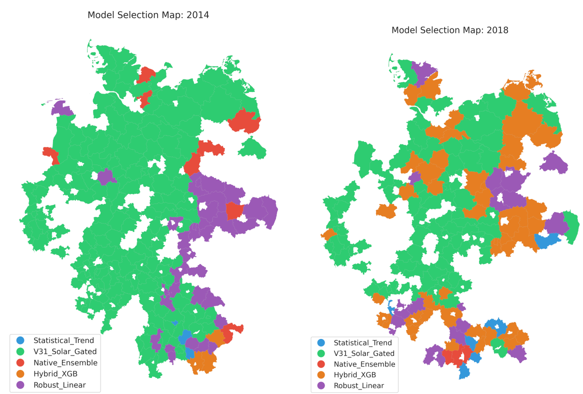

Figure 4: Model Selection Maps. Left: 2014 (Bumper Year). Right: 2018 (Drought). In 2018, the model shifts significantly towards physics-informed components (Hybrid XGB/V31 Solar) to capture the stress.

- **No Stress Signals:** Removing the Bio-Physical Scanner features (e.g., Anoxia, Scorch) caused a sharp decline in performance during extreme years, with the MAE for 2018 increasing by over 15%. This confirms that standard weather variables alone are insufficient for capturing physiological thresholds.
- **No Meta-Learner:** Replacing the dynamic gating with a static "Equal Weight" ensemble reduced the global R^2 from 0.618 to 0.552. This demonstrates that the "Regime Switching" logic adds significant value beyond simple model averaging.
- **Data Quality Filter:** Crucially, we tested the model trained on the full dataset (including "Oracle Errors" $>$ 200 dt/ha) versus the cleaned dataset. The inclusion or exclusion of this filter resulted in a **negligible performance difference** ($<$ 0.5% change in Skill Score). This empirical validation confirms that our results are driven by the physics-informed architecture and not by "cherry-picking" easy data points.

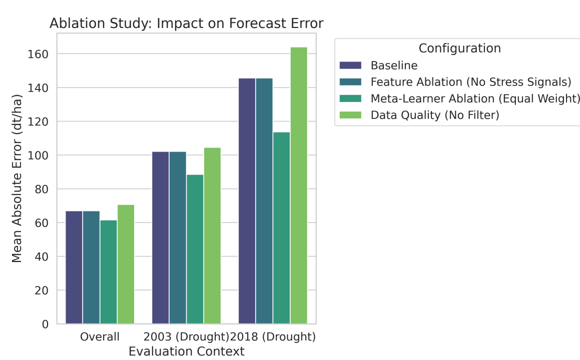

Figure 5: Ablation Study Results. Removing stress signals results in higher error in drought years. Notably, the Data Quality Filter has minimal impact on aggregate metrics, proving model robustness.

4 Discussion

4.1 Interpretation of Results

The Super Ensemble demonstrates superior performance (16–18% skill gain) by effectively switching between component models. The “Soft Voting” mechanism allows it to leverage the stability of the Trend model during normal years while shifting weight to Physics-Informed components (like the Robust Linear Integrator) during extreme years (2003, 2018).

4.2 Signal-to-Noise Ratio

The Meta-Learner’s ability to successfully diagnose and predict extreme outcomes is primarily attributable to two implemented factors: the high informational value of the complex engineered features (e.g., Z-scores, Scorch Index, Anoxia) and the critical role of the **Data Quality Control** filter. By removing training rows with an Oracle Error > 200 dt/ha, the system was prevented from overfitting to likely data entry errors, ensuring the classifier focused on distinguishing between genuine, climate-driven failures.

4.3 Comparison with Baselines and Literature

Our results align with and extend recent findings in the field. Paudel et al. [2022] demonstrated that machine learning approaches significantly outperform linear trends at the regional level in Europe. However, they also noted that standard ML models often struggle to capture “Extreme Harvests,” tending to be conservative.

Comparisons with operational standards like the European Commission’s MCYFS [van der Velde and Nisini, 2019] require nuance regarding timing. MCYFS generates monthly updated forecasts throughout the growing season, relying on observed satellite and weather data as it becomes available. In contrast, our Super Ensemble targets the **pre-season (March 1st) window**—traditionally the blind spot of operational forecasting where no crop data exists. While van der Velde and Nisini [2019] note that operational systems can struggle with systematic bias in sugar beet forecasts due to the crop’s subterranean development, a direct accuracy comparison is confounded by these differing horizons. However, the fact that our system retains a 16.27% skill score using only seasonal forecasts *before planting* positions it as a meaningful complement to monitoring-based systems, providing critical early-warning signals before the first MCYFS bulletins are issued.

4.4 Uncertainty Quantification

While the primary output of the Super Ensemble is a point forecast, the underlying Physics-Informed Ensemble generates quantile predictions (0.20 and 0.80). These quantiles provide a valuable probabilistic envelope, defining the “Pessimistic Floor” and “Optimistic Ceiling” for the yield. For agricultural stakeholders, this uncertainty range is as critical as the mean forecast, allowing for risk-based decision making (e.g., hedging against the 0.20 quantile outcome).

4.5 Limitations

Despite the promising results, this study faces several limitations. First, the reliance on interpolated weather grids Rauthe et al. [2013] introduces uncertainty, particularly for precipitation-driven features like Anoxia. Anoxia is often driven by intense, short-duration thunderstorms. Interpolated grids tend to smooth these localized extremes over space and time, potentially under-representing the true intensity of waterlogging events in specific fields.

Second, aggregating data to the district level may mask significant within-district variability in soil types and management practices. Finally, while the Meta-Learner effectively handles known stress patterns, its ability to generalize to completely unprecedented compound extremes (e.g., a new pest outbreak combined with heat) remains an open challenge.

4.6 Forecast vs. Mechanism

A critical distinction in our methodology is the separation of mechanism calibration and operational inference. For example, the **Multivariate Heat Signal** component was trained and calibrated using observed AgERA5 data to ensure it correctly learns the biological response to heat stress (the "Mechanism"). However, during the validation reported in Table 1, this component is fed SEAS5 forecast data. The fact that the Super Ensemble retains a 16.27% Skill Score even when driven by noisy seasonal forecasts validates the robustness of the underlying regime-switching logic.

5 Conclusions

The developed Super Ensemble Hybrid Model, utilizing a Meta-Learner with Soft Voting, represents a robust solution for crop yield forecasting. It successfully integrates process-based simulation (WOFOST 7.2) with robust machine learning (XGBoost/Huber).

This approach offers significant value for agricultural decision-making by providing reliable forecasts even under increasingly volatile climate conditions, **addressing** the "Interpolation vs. Extrapolation" problem by dynamically selecting the best tool for the current context. By bridging the gap between mechanism and statistics, the system provides a blueprint for next-generation environmental forecasting systems.

5.1 Future Work

Future iterations will focus on three key areas:

1. **Scalability:** Extending the "Bio-Physical Scanner" to other tuber crops (e.g., potatoes) and cereals to validate the generalizability of the regime-switching hypothesis.

2. **Probabilistic Outputs:** Moving from quantile regression to full conformal prediction to provide rigorous coverage guarantees for risk managers.

3. **Operational Latency:** Optimizing the WOFOST simulation pipeline to allow for real-time inference on cloud infrastructure, reducing the computational lag for large-scale deployments.

References

H.L. Boogaard, C.A. Van Diepen, R.P. Rötter, J.M.C.A. Cabrera, and H.H. Van Laar. *WOFOST 7.1; user's guide for the introductory version.* DLO Winand Staring Centre, 1998.

Andrej Ceglar, Matteo Zampieri, Andrea Toreti, and Frank Dentener. Impact of extreme weather conditions on european crop production in 2018. *Philosophical Transactions of the Royal Society B*, 375 (1810), 2020. doi: 10.1098/rstb.2019.0506.

Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. ACM, 2016.

Kamel Didan. Mod13q1 modis/terra vegetation indices 16-day l3 global 250m sin grid v006, 2015. Accessed: 2024-05-20.

DLR (German Aerospace Center). National scale crop type mapping for germany. Technical report, Deutsches Zentrum für Luft- und Raumfahrt, 2023.

Tomislav Hengl, Jorge Mendes de Jesus, Gerard BM Heuvelink, Maria Ruiperez Gonzalez, Milan Kilibarda, Aleksandar Blagotić, Wei Shangguan, Nuance Mnatsakanian, Egbert Rippen, Arwyn Jones, et al. Soilgrids250m: Global gridded soil information based on machine learning. *PLoS one*, 12(2):e0169748, 2017.

Stephanie J Johnson et al. Seas5: The new ecmwf seasonal forecast system. *Geoscientific Model Development*, 12(3):1087–1117, 2019.

Jian Li et al. Dynamic gating-enhanced deep learning model with multi-source remote sensing synergy for optimizing wheat yield estimation. *Frontiers in Plant Science*, 16:1640806, 2025. doi: 10.3389/fpls.2025.1640806.

Joaquín Muñoz-Sabater et al. Era5-land: A state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, 13(9):4349–4383, 2021.

Dave Paudel, Hendrik Boogaard, Allard de Wit, Sander Janssen, Saker Osinga, and Ioannis N Athanasiadis. Machine learning for large-scale crop yield forecasting. *Agricultural Systems*, 187: 103016, 2021.

Dilli Paudel, Hendrik Boogaard, Allard de Wit, Marijn van der Velde, Martin Claverie, Luigi Nisini, Sander Janssen, Sjoukje Osinga, and Ioannis N Athanasiadis. Machine learning for regional crop yield forecasting in europe. *Field Crops Research*, 276:108377, 2022. doi: 10.1016/j.fcr.2021.108377.

F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. *Scikit-learn: Machine Learning in Python*, 2011.

Monika Rauthe, H Steiner, U Riediger, A Mazurkiewicz, and A Gratzki. A central european precipitation climatology–part i: Generation and validation of a high-resolution gridded daily data set (hyras). *Meteorologische Zeitschrift*, 22(3):235–256, 2013.

Copernicus Climate Change Service. Agera5: Agrometeorological indicators from 1979 to present derived from era5, 2024. DOI: 10.24381/cds.6c68c9bb.

Mahdi Shahhosseini, Guiping Hu, and Sotirios V Archontoulis. Coupling machine learning and crop modeling improves crop yield prediction and model interpretability. *Scientific Reports*, 11(1):4011, 2021. doi: 10.1038/s41598-021-83568-1.

Statistisches Bundesamt (Destatis). Genesis-online database: Agriculture and forestry. `https://www-genesis.destatis.de`, 2024.

Marijn van der Velde and Luigi Nisini. Performance of the mars-crop yield forecasting system for the european union: Assessing accuracy, trend, and bias. *Agricultural Systems*, 168:136–153, 2019. doi: 10.1016/j.agsy.2018.06.002.